



Answer key: EDA on the diamonds dataset

Exam Proctor faculty pilot · Monday 5 October 2026 · Part B, 11:38 to 12:15 · Old IQAC · 40 marks

For organisers and invigilators only

Do not attach this to the participants' email. Every value below was computed by running the code shown on the seaborn `diamonds` dataset (seaborn 0.13.2, pandas 3.0.6).

First cell

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = sns.load_dataset('diamonds') # 53,940 diamonds, built into seaborn
df.head()
```

Question 1

2 marks

Load the diamonds dataset and report the number of rows and columns.

```
df.shape
```

Answer. 53,940 rows and 10 columns.

Question 2

3 marks

Show the data type of every column. Which columns are categorical?

```
df.dtypes
```

Answer. Categorical: cut, color, clarity. Numeric: carat, depth, table, x, y, z (float) and price (integer).

Question 3

3 marks

How many missing values are there in the whole dataset? How many rows are exact duplicates?

```
print('Missing values:', df.isna().sum().sum())
print('Duplicate rows:', df.duplicated().sum())
```

Answer. 0 missing values. 146 duplicate rows.

Question 4**4 marks**

For `price` and `carat`, report the mean, median, minimum and maximum. Is `price` right-skewed? Justify using the mean and the median.

```
df[['price', 'carat']].agg(['mean', 'median', 'min', 'max']).round(2)
```

Answer. Price: mean 3,932.80, median 2,401, minimum 326, maximum 18,823. Carat: mean 0.80, median 0.70, minimum 0.20, maximum 5.01. Price is right-skewed: its mean is well above its median (skewness 1.62), because a few very expensive diamonds pull the mean up.

Question 5**4 marks**

How many diamonds are there of each `cut`? Which cut is the most common, and what percentage of all diamonds is it?

```
counts = df['cut'].value_counts()
print(counts)
print('Share of the most common cut:', round(100 * counts.iloc[0] / len(df), 1),
      '%')
```

Answer. Ideal 21,551; Premium 13,791; Very Good 12,082; Good 4,906; Fair 1,610. Ideal is the most common, at 40.0% of all diamonds.

Question 6**6 marks**

Compute the mean price for each `cut`, sorted from highest to lowest. The best cut (Ideal) has the lowest mean price. Compute the mean `carat` for each cut and use it to explain why.

```
print(df.groupby('cut', observed=True)
      ['price'].mean().sort_values(ascending=False).round(0))
print(df.groupby('cut', observed=True)['carat'].mean().round(2))
```

Answer. Mean price: Premium 4,584; Fair 4,359; Very Good 3,982; Good 3,929; Ideal 3,458. Mean carat: Fair 1.05; Premium 0.89; Good 0.85; Very Good 0.81; Ideal 0.70. Ideal diamonds are the smallest on average, and size (carat) drives price far more than cut, so comparing cuts without allowing for carat is misleading (a confounding variable).

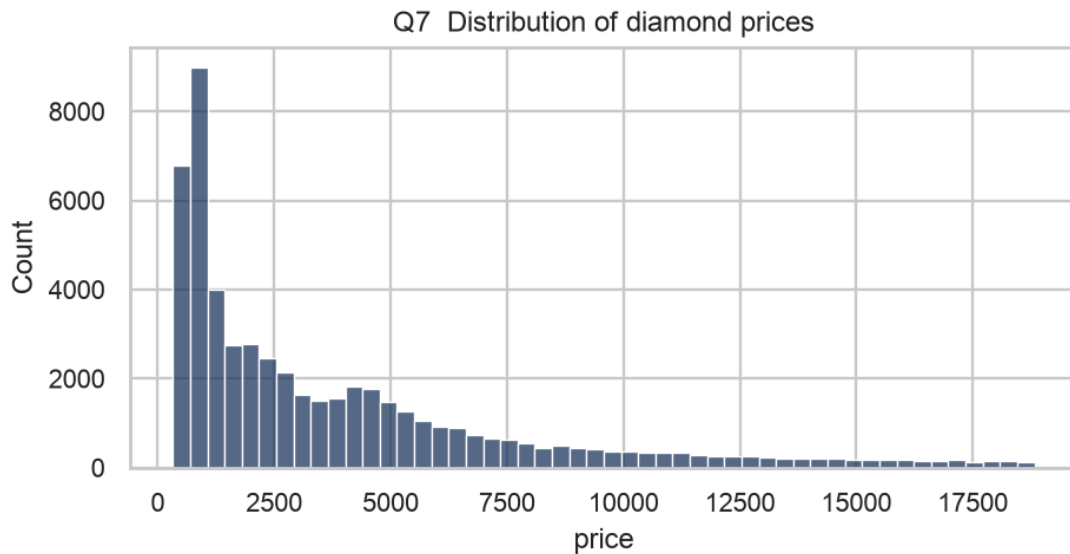
Question 7

4 marks

Plot a histogram of `price`. Describe its shape in one sentence.

```
sns.histplot(data=df, x='price', bins=50)
plt.title('Distribution of diamond prices')
plt.show()
```

Answer. Strongly right-skewed: most diamonds cost under about 5,000 (75% cost 5,324 or less), with a long tail up to 18,823.



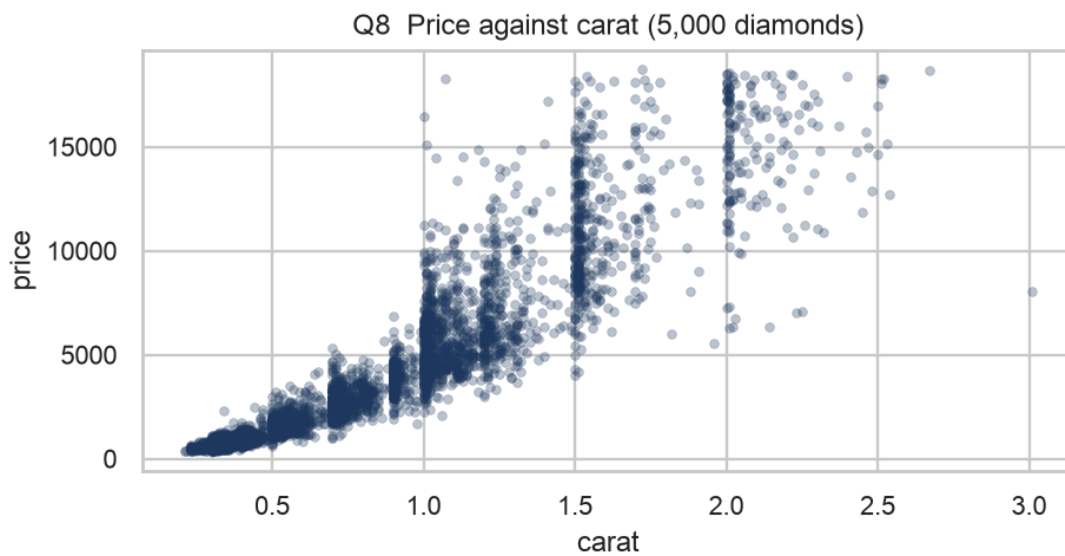
Question 8

4 marks

Draw a scatter plot of `carat` (x) against `price` (y). Use transparency or a random sample of 5,000 rows so the points are readable. Describe the relationship.

```
sample = df.sample(5000, random_state=1)
sns.scatterplot(data=sample, x='carat', y='price', alpha=0.3)
plt.title('Price against carat (5,000 diamonds)')
plt.show()
```

Answer. A strong, positive, non-linear relationship: price rises faster than carat, and the spread of prices widens for larger stones. Points cluster at popular sizes such as 0.3, 0.7 and 1.0 carat.



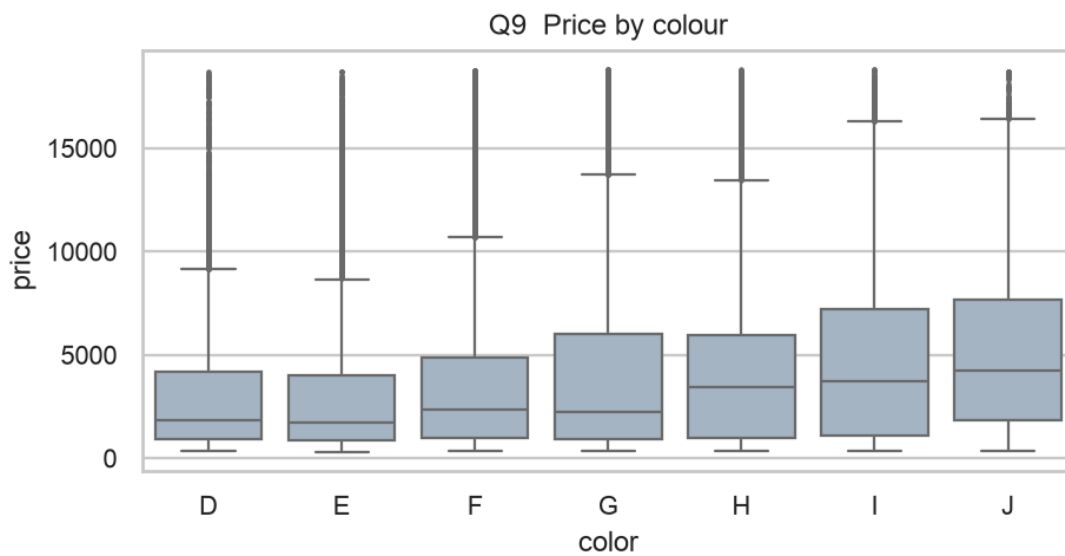
Question 9

4 marks

Draw a box plot of `price` for each `color`. Which colour has the highest median price? (D is the best colour and J the worst.)

```
sns.boxplot(data=df, x='color', y='price')
plt.title('Price by colour')
plt.show()
print(df.groupby('color', observed=True)
      ['price'].median().sort_values(ascending=False))
```

Answer. J has the highest median price (4,234), then I (3,730) and H (3,460); D (1,838) and E (1,739) are the lowest. Again a size effect: J stones tend to be larger.



Question 10**6 marks**

Compute the correlation matrix of the numeric columns. Which variable is most strongly correlated with price, and what is the correlation? (Optional: show it as a heatmap.)

```
corr = df.select_dtypes('number').corr()
print(corr['price'].sort_values(ascending=False).round(3))
sns.heatmap(corr, annot=True, fmt='.2f', cmap='Blues')
plt.show()
```

Answer. Carat is the most strongly correlated with price ($r = 0.922$), followed by the dimensions x (0.884), y (0.865) and z (0.861). Depth (-0.011) and table (0.127) are barely related to price.

